

Beumsu Kim

AI Systems Engineer

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Github

<https://beusmkim.github.io/profile/>



AI Systems Engineer specializing in diagnosing and optimizing inference bottlenecks under real GPU constraints. Profiled transformer-based workloads on NVIDIA A100, identified memory-bandwidth-bound behavior, and restructured execution paths using precision strategies (FP16/INT8) and architectural isolation to deliver measurable performance gains.



Core Technical Skills

- GPU Profiling: torch.profiler, nvidia-smi, roofline-based performance interpretation
- Precision Optimization: FP16 mixed precision, INT8 quantization trade-off analysis
- Runtime Design: CUDA stream concepts, compute isolation (MSA), concurrency stabilization
- Transformer Analysis: Attention memory footprint and bandwidth behavior diagnosis
- Distributed Systems: AWS Batch, S3, file-based partitioned execution



Experience

**Jul 2025 -
Jun 2026**

AI Systems / Inference Optimization Engineer

Samsung Software AI Academy For Youth, Gwangju, South Korea

- Profiled transformer-based inference on NVIDIA A100 using torch.profiler and runtime memory monitoring
- Identified memory-bandwidth-bound behavior before peak SM utilization
- Reduced peak VRAM by ~38–40% via FP16 mixed precision deployment
- Improved end-to-end inference latency by ~22% through execution path restructuring
- Designed compute isolation architecture (MSA) reducing blocking by ~35% under concurrency
- Implemented multi-pass inference orchestration improving accuracy from 92% → 96.7%
- Engineering focus: profiling-driven diagnosis and architectural isolation for reproducible gains

Mar 2021 - Feb 2024 Sales Operations Manager

LG U+ Retail, Gyeonggi-do, South Korea

- Redesigned store flow and structured CRM consultation process
- Improved accessory attachment rate and operational consistency
- Learned bottleneck isolation and flow optimization under real-world constraints

2021 - 2021 Fashion Brand (Founder / Operator)

Illuminate — Fashion Brand, Seoul, South Korea

- Launched and operated online fashion brand
- Managed sourcing, pricing, marketing, and fulfillment
- Applied data-driven iteration based on conversion and traffic analysis

Education

Aug 2025 Bachelor of Science: Mechanical Engineering

Kyung Hee University

- Constraint-driven problem solving (thermodynamics, fluid mechanics, materials)
- Systems-level reasoning under physical and resource limitations

Projects — Details in my Github.

AI Image Analysis — Memory-Bound Inference Optimization

- Diagnosed bandwidth-bound behavior on NVIDIA A100
- Reduced peak VRAM by ~38–40% via FP16 deployment
- Improved end-to-end latency by ~22% through execution restructuring

OneTakeStudio — Compute-Isolated AI Inference System

- Decoupled streaming and inference workers
- Reduced blocking by ~35% under concurrent workloads

High-Strain-Rate Material Characterization (SHPB)

- Calibrated constitutive model using empirical SHPB data
- Reduced simulation error via iterative validation in Abaqus

Software

Python

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Advanced

Git

◆◆◆◆◆
Advanced

FastAPI

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Advanced